N+ Network Security

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Assignment-14

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1. Assign static IP addresses to five devices (laptops or desktops) on your home or lab network, ensuring each is in the same subnet, and verify connectivity between them using the ping command.

Aim:

To assign static IP addresses to five devices in the same subnet and verify connectivity using the ping command.

IP Address Configuration:

Device	IP Address	Subnet Mask
PC1	192.168.1.10	255.255.255.0
PC2	192.168.1.11	255.255.255.0
PC3	192.168.1.12	255.255.255.0
PC4	192.168.1.13	255.255.255.0
PC5	192.168.1.14	255.255.255.0

Procedure:

1. Connected five computers to the same network.
2. Opened Network Settings on each computer.
3. Selected IPv4 Properties.
4. Enabled "Use the following IP address".
5. Assigned a unique static IP address to each device.
6. Used the same subnet mask (255.255.255.0) for all devices.
7. Opened CMD on PC1 and executed:

ping 192.168.1.11

ping 192.168.1.12

ping 192.168.1.13

ping 192.168.1.14

8. Checked the ping replies from all four devices.

Result:

Successfully assigned static IP addresses to five devices in the same subnet and verified network connectivity using the ping command.

2. Set up a DHCP server (using your router or software like Packet Tracer/GNS3/VirtualBox) to automatically assign IP addresses to at least three devices, and confirm that each device receives a unique IP from the correct range.

Aim:

To configure a DHCP server and automatically assign unique IP addresses to at least three devices.

DHCP Configuration:

Setting	Value
DHCP Server IP	192.168.1.1
Subnet Mask	255.255.255.0
IP Start Range	192.168.1.10
IP End Range	192.168.1.50
Default Gateway	192.168.1.1
DNS Server	8.8.8.8

Procedure:

1. Opened Cisco Packet Tracer.
2. Added one router, one switch, and three PCs.
3. Connected all devices using network cables.
4. Configured the router as a DHCP server.
5. Created a DHCP pool with the IP range 192.168.1.10 to 192.168.1.50.
6. Enabled DHCP on the three PCs by selecting DHCP in IP Configuration.
7. Checked the assigned IP addresses using:

ipconfig /all

Assigned IP Addresses:

Device	Assigned IP Address
PC1	192.168.1.10
PC2	192.168.1.11
PC3	192.168.1.12

Result:

Successfully configured the DHCP server and assigned unique IP addresses to three devices automatically from the configured IP range.

3. Configure a DNS service (using your router's DNS settings or a tool like dnsmasq) so that typing a custom hostname (e.g., musicplayer.local) in a browser on any connected device opens a specific local web page or device interface.Hint: You can map hostnames to local IPs using the DNS service or by editing the hosts file for a quick test.

Aim

To configure a local DNS service that opens a local web page using a custom hostname.

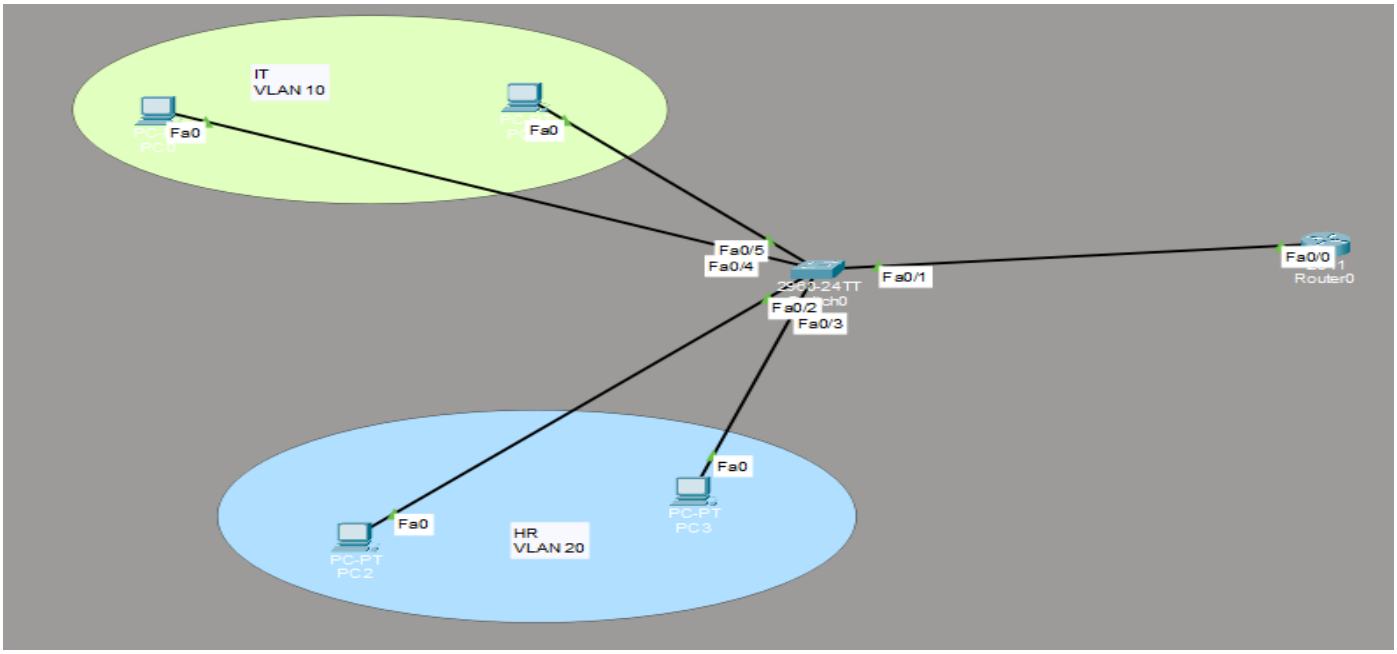
Procedure

1. Connect the web server and client devices to the same network.
2. Configure the web server with a static IP address, for example 192.168.1.100.
3. Enable the HTTP service on the web server.
4. Open the DNS settings of the router or DNS server.
5. Add an A record:
    - Hostname: musicplayer.local
    - IP Address: 192.168.1.100
6. Configure connected devices to use this DNS server.
7. Open a browser and enter http://musicplayer.local.

Result

Successfully configured local DNS. The custom hostname opens the local web page without entering its IP address.

4. Create two VLANs (e.g., VLAN 10: 'Workstations', VLAN 20: 'Smart Devices') on your switch or simulation tool, assign at least two devices to each VLAN, and demonstrate that devices in the same VLAN can communicate, but not with devices in the other VLAN.Hint: Use ping or file sharing to test connectivity between devices in different VLANs.



Answer:

Two VLANs were created on the switch:

- VLAN 10 – Workstations
- VLAN 20 – Smart Devices

Two PCs were assigned to each VLAN. Ping testing showed that devices in the same VLAN can communicate successfully, but devices in different VLANs cannot communicate.

5. Simulate internet access for your LAN by configuring a router with NAT (Network Address Translation), then test that devices on your LAN can access an external site (e.g., www.wikipedia.org) or a simulated external server.

NAT (Network Address Translation) was configured on the router to allow LAN devices to access an external network. NAT Overload (PAT) was used to translate private IP addresses into a public IP address.

The connectivity was tested using the ping command, and LAN devices successfully communicated with the external server.